



Introduction to HPC Clusters

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Agenda

- HPC Clusters

- Leonardo Overview

- Leonardo Deep Dive (Jupyter notebook)

- Demo

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About Andrea

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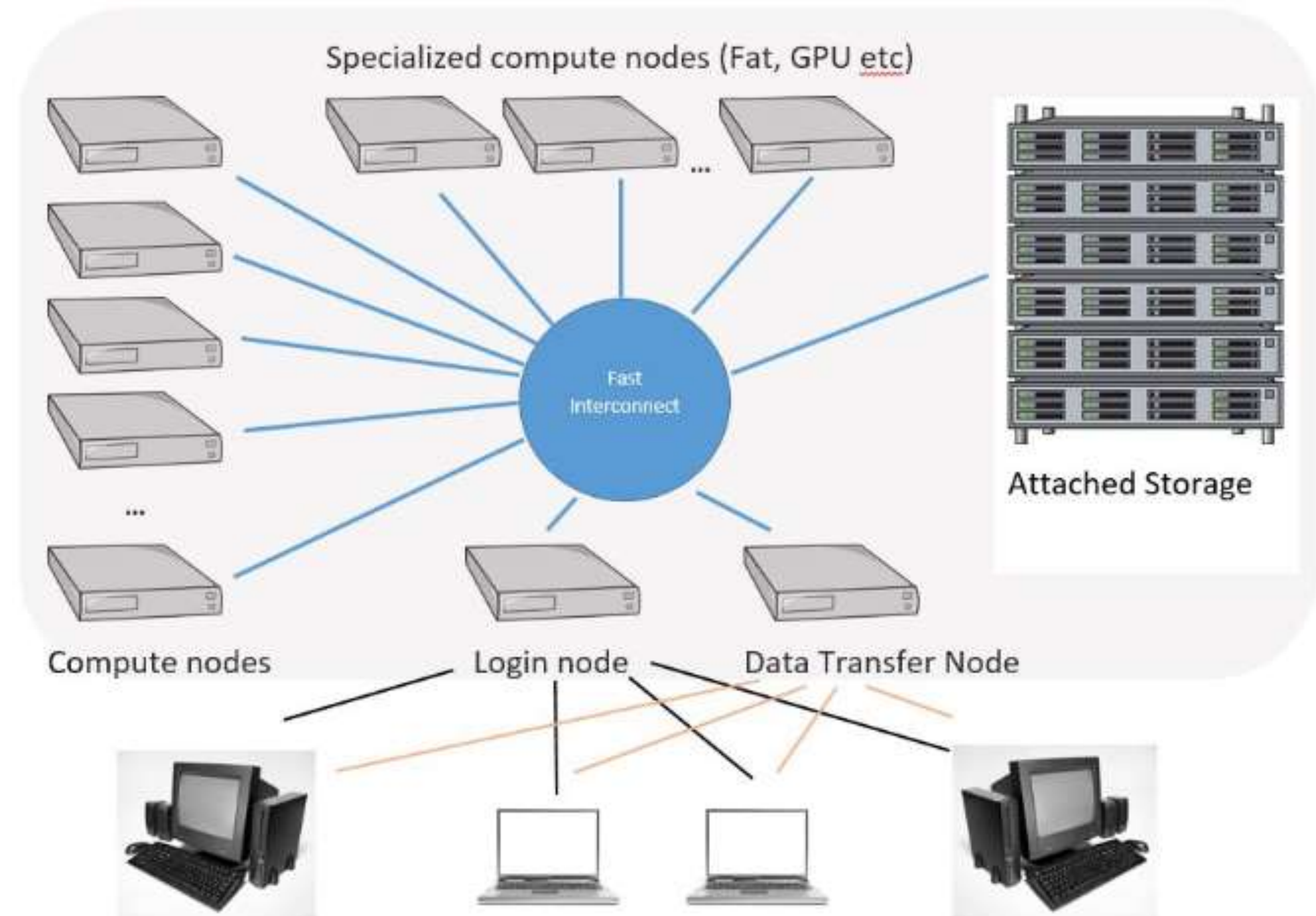
- Since May 2022 Solution Architect at NVIDIA
 - Supporting HER through NVAITC
- Postdoc at Aalto (Helsinki, Finland, 2021-2022)
 - Uncertainty quantification for deep learning
- Previous industrial experience (2019-2021)
 - Domain Adaptation, Huawei Ireland, Dublin
- Ph.D. in CS @ Trento (2016-2019)
 - Supervised by Elisa Ricci & Nicu Sebe
 - Main topic stereo matching



HPC Cluster

What is an HPC Cluster?

- An HPC cluster is a collection of many separate servers (computers), called nodes, which are connected via a fast interconnect
- A HPC cluster typically has
 - a headnode or login node, where users log in
 - a specialized data transfer node
 - regular compute nodes (where majority of computations is run)
 - GPU nodes (on these nodes computations can be run both on CPU cores and on a Graphical Processing Unit)
 - an Infiniband switch to connect all nodes
- All cluster nodes have the same components as a laptop or desktop: CPU cores, memory and disk space. The difference between personal computer and a cluster node is in quantity, quality and power of the components.



Credits: <https://www.hpc.iastate.edu/guides/introduction-to-hpc-clusters/what-is-an-hpc-cluster>

HPC Clusters Top500

<https://top500.org/lists/top500/2024/06/>

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	Frontier - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE DOE/SC/Oak Ridge National Laboratory United States	8,699,904	1,206.00	1,714.81	22,786
2	Aurora - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
3	Eagle - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	
4	Supercomputer Fugaku - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
5	LUMI - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE EuroHPC/CSC Finland	2,752,704	379.70	531.51	7,107
6	Alps - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	1,305,600	270.00	353.75	5,194
7	Leonardo - BullSequana XH2000, Xeon Platinum 8358 32C 2.6GHz, NVIDIA A100 SXM4 64 GB, Quad-rail NVIDIA HDR100 Infiniband, EVIDEN EuroHPC/CINECA	1,824,768	241.20	306.31	7,494

Workload Management

Slurm

- HPC clusters need workload management software to match jobs and available resources
- SLURM is one of the most widely utilized
- Key features are
 - Allocation of resources
 - Environment suitable for distributed jobs (MPI)
 - Management of the job queues
- Lots of features can be implemented
 - Queue priorities
 - Reservations
 - Partitions (GPU/CPU)
 - Dynamic priority based on computational budgeted consumption

Software

- Typically a cluster has some software installed that we can just load and use.
 - E.g. CINECA has modules installed
- Users can compile their code and install additional software

Data Transfer – Internet Access

- Login nodes are the only ones connected to internet, compute nodes are in a separate network
- Small data can be downloaded from the login node
- Big data is downloaded through “data mover” nodes
- Not all applications need lots of data, AI is very data hungry!



Leonardo Overview

Leonardo CINECA

Specs

	Booster	DCGP
Model	Atos BullSequana X2135 "Da Vinci" single-node GPU blade	Atos BullSequana X2140 three-node CPU blade
Racks	116	22
Nodes	3456	1536
Processors	single socket 32 cores Intel Ice Lake CPU 1 x Intel Xeon Platinum 8358, 2.60GHz TDP 250W	dual socket 56 cores Intel Sapphire Rapids CPU 2 x Intel Xeon Platinum 8480p, 2.00 GHz TDP 350W
Accelerators	4 x NVIDIA Ampere GPUs/node, 64GB HBM2e NVLink 3.0 (200GB/s)	-
Cores	32 cores/node	112 cores/node
RAM	512 (8x64) GB DDR4 3200 MHz	512 (16 x 32) GB DDR5 4800 MHz
Peak Performance	about 309 Pflop/s	9 Pflops/s
Internal Network	DragonFly+ 200 Gbps (NVIDIA Mellanox Infiniband HDR)	
	2 x dual port HDR100 per node	single port HDR100 per node
Storage (raw capacity)	137.6 PB based on DDN ES7990X and Hard Drive Disks (Capacity Tier) 5.7 PB based on DDN ES400NVX2 and Solid State Drives (Fast Tier)	



Leonardo CINECA

Available Memory

	Total Dimension (TB)	Quota (GB)	Notes
\$HOME	0.46 PiB	50GB per user	• permanent • backed up • user specific
\$CINECA_SCRATCH	40 PiB	no quota	• HDD storage • temporary • user specific • no backup • automatic cleaning procedure of data older than 40 days (time interval can be reduced in case of critical usage ratio of the area. In this case, users will be notified via HPC-News).
\$PUBLIC	0.46 PiB	50GB per user	• permanent • user specific • no backup
\$WORK	30 PB	1TB per project	• permanent • project specific • no backup • extensions can be considered if needed (mailto: superc@cineca.it)
\$FAST	3.5PB	1TB per project	• permanent • project specific • no backup

Leonardo CINECA

Queues

SLURM partition	Job QOS	# cores/# GPU per job	max walltime	max running jobs per user/ max n. of nodes/cores/GPUs per user	priority	notes
lrd_all_serial (default)	<i>normal</i>	max = 4 physical cores (8 logical cpus) max mem = 30800 MB	04:00:00	1 node / 4 cores / 30800 MB	40	No GPUs Hyperthreading x2
boost_usr_prod	<i>normal</i>	max = 64 nodes	24:00:00		40	
	boost_qos_dbg	max = 2 nodes	00:30:00	2 nodes / 64 cores / 8 GPUs	80	
	boost_qos_bprod	min = 65 nodes max = 256 nodes	24:00:00	256 nodes	60	runs on 1536 nodes min is 65 FULL nodes
	boost_qos_lprod	max = 3 nodes	4-00:00:00	3 nodes / 12 GPUs	40	

Leonardo CINECA

Software

- CINECA HPC department installed for us already a lot of software
- They use modules, grouped in profiles
 - There are chemistry, physics, deep learning, statistics, etc. profiles
 - Each profile contains the most important software for each application
- There are base modules common to every profile (e.g. compilers)
- The deep learning profiles also comes with some widely used datasets (e.g. imagenet)

Leonardo Deep Dive ([Jupyter notebook link](#))



Demo

Demo topics

Connect to the cluster

- `ssh user@login.leonardo.cineca.it`
- The welcome messages tells you details about the cluster and most of the time will remind you about upcoming maintenance of the cluster

```
Welcome to:

      Leonardo

*****
* Red Hat Enterprise Linux 8.7 (Ootpa)                                *
*                                                                 *
* Booster module:                                                  *
* Atos Bull Sequana X2135 "Da Vinci" Blade                         *
* 3456 compute nodes with:                                         *
*   - 32 cores Ice Lake at 2.60 GHz                               *
*   - 4 x NVIDIA Ampere A100 GPUs, 64GB                           *
*   - 512 GB RAM                                                  *
*                                                                 *
* DataCentric General Purpose module (DCGP):                      *
* Atos BullSequana X2140 Blade                                     *
* 1536 compute nodes with:                                         *
*   - 2x56 cores Intel Sapphire Rapids at 2.00 GHz               *
*   - 512 GB RAM                                                  *
*                                                                 *
* Internal Network: 200G HDR Infiniband Dragonfly+                *
* SLURM 22.05                                                       *
*                                                                 *
* For a guide on Leonardo:                                         *
* https://wiki.u-gov.it/confluence/display/SCAIUS/UG3.2%3A+LEONARDO+UserGuide *
* For support: superc@cinca.it                                     *
*****
IN EVIDENCE:
- A new personal area $PUBLIC is available to share installations and/or
  data. Please, keep in mind that the $PUBLIC directory is by default open
  to everybody on the cluster, and your files are visible to all users.
- The automatic cleaning of the $SCRATCH area is NOT active at the moment
- RCM will be available soon
- Spack module is available to customize your software environment.
  "module av spack" to list the available versions and
  "module load spack/<version>" to use a specific one
=====
```


Demo topics

Check the cluster (sinfo)

- Get information on the cluster **sinfo -s**
- Output is huge because the cluster is huge
- You can see different partitions, booster is the GPU one with 3500 nodes, datacentric is the CPU partition with 1500 nodes
- You can see A active nodes, I idle nodes, O off nodes, T total number of nodes

```
[apilzer0@login07 ~]$ sinfo -s
PARTITION      AVAIL  TIMELIMIT  NODES(A/I/O/T)  NODELIST
boost_usr_prod  up 1-00:00:00  3195/58/5/3258  lrdn[0001-0039,0041-0043,0045-0094,0096-0150,0152-0173,0175-0178,0180-0189,0191-0243,0245-0250,0252-0253,025
5,0258-0270,0272-0287,0289-0294,0296-0317,0319-0351,0353,0355-0360,0362,0364-0373,0375-0380,0382-0389,0391-0455,0457-0464,0466-0469,0471-0489,0491-0510,0512-0
522,0524-0538,0540-0562,0564-0621,0623-0667,0669-0670,0672-0685,0687-0709,0711-0763,0765-0829,0831-0838,0840-0884,0886-0896,0898-0936,0938-0945,0947-0969,0971
-0975,0977-0998,1000,1002-1003,1005-1012,1015,1017-1036,1039-1108,1110-1115,1117-1128,1130-1139,1141-1153,1155-1163,1165,1167-1184,1186-1187,1189-1205,1207-12
08,1210-1235,1237-1419,1421-1426,1428-1456,1458-1473,1475-1540,1542-1545,1547-1577,1579-1589,1591-1638,1640-1660,1662,1664-1668,1671-1695,1697-1709,1711-1724,
1726-1751,1753-1754,1756-1770,1772-1773,1775-1783,1785-1797,1799-1820,1822-1824,1826-1844,1846-1853,1855-1899,1901-1947,1949-1961,1963-2012,2014-2019,2021-202
7,2029-2066,2068-2070,2072-2084,2086-2102,2104-2124,2126-2138,2140-2147,2149-2170,2172-2179,2181-2217,2219-2343,2345-2348,2350-2356,2359-2362,2365-2367,2369-2
383,2385-2387,2389-2396,2398-2421,2423-2437,2439-2490,2492-2499,2501-2524,2526-2539,2541,2543-2550,2552-2561,2564-2571,2573-2574,2576-2581,2584-2592,2594-2597
,2599-2600,2602-2607,2609-2618,2620-2628,2630-2653,2655-2662,2664-2696,2698-2703,2705-2733,2735-2738,2740-2754,2756-2769,2771-2774,2776-2800,2802-2812,2814-28
57,2859-2864,2866-2873,2875-2902,2904-2911,2914-2926,2928-2974,2977-2988,2990-2992,2994-3011,3014-3023,3025-3055,3057-3068,3070-3090,3092-3098,3100-3108,3110,
3112-3120,3122-3124,3126-3129,3131-3133,3135-3142,3144-3150,3152-3177,3179,3181-3186,3188-3202,3204-3212,3214-3220,3222-3225,3227-3241,3243-3278,3280-3281,328
3-3284,3286-3289,3291-3297,3299-3301,3303-3313,3315-3344,3346-3354,3357-3394,3396-3408,3410-3415,3417-3423,3425-3429,3431-3447,3449-3456]
dcgp_usr_prod   up 1-00:00:00  1021/424/0/1445  lrdn[3457-3459,3461-3478,3480,3482-3492,3498-3513,3515-3567,3569-3576,3580-3612,3614-3651,3653-3657,3659-367
1,3673-3721,3724-3751,3753-3701,3799-3808,3810-3831,3833-3843,3845-3861,3863-3906,3908-3915,3917-3936,3940-3941,3943,3945-3951,3953-3977,3979-3989,3992-4032,4
034-4054,4056-4130,4132-4134,4137-4140,4142-4152,4154-4176,4178-4215,4218-4232,4234-4343,4345-4346,4348-4382,4384-4416,4418-4425,4427-4448,4451-4458,4460-4472
,4474-4488,4492-4494,4496-4524,4526-4542,4544-4593,4595-4609,4611-4662,4664-4668,4672-4675,4677-4681,4683-4698,4700-4705,4707-4716,4718-4767,4769-4770,4772-47
85,4787-4788,4790-4818,4820-4861,4863-4886,4890-4967,4970-4989,4991-4992]
boost_fua_prod  up 1-00:00:00  3195/58/5/3258  lrdn[0001-0039,0041-0043,0045-0094,0096-0150,0152-0173,0175-0178,0180-0189,0191-0243,0245-0250,0252-0253,025
5,0258-0270,0272-0287,0289-0294,0296-0317,0319-0351,0353,0355-0360,0362,0364-0373,0375-0380,0382-0389,0391-0455,0457-0464,0466-0469,0471-0489,0491-0510,0512-0
522,0524-0538,0540-0562,0564-0621,0623-0667,0669-0670,0672-0685,0687-0709,0711-0763,0765-0829,0831-0838,0840-0884,0886-0896,0898-0936,0938-0945,0947-0969,0971
-0975,0977-0998,1000,1002-1003,1005-1012,1015,1017-1036,1039-1108,1110-1115,1117-1128,1130-1139,1141-1153,1155-1163,1165,1167-1184,1186-1187,1189-1205,1207-12
08,1210-1235,1237-1419,1421-1426,1428-1456,1458-1473,1475-1540,1542-1545,1547-1577,1579-1589,1591-1638,1640-1660,1662,1664-1668,1671-1695,1697-1709,1711-1724,
1726-1751,1753-1754,1756-1770,1772-1773,1775-1783,1785-1797,1799-1820,1822-1824,1826-1844,1846-1853,1855-1899,1901-1947,1949-1961,1963-2012,2014-2019,2021-202
7,2029-2066,2068-2070,2072-2084,2086-2102,2104-2124,2126-2138,2140-2147,2149-2170,2172-2179,2181-2217,2219-2343,2345-2348,2350-2356,2359-2362,2365-2367,2369-2
383,2385-2387,2389-2396,2398-2421,2423-2437,2439-2490,2492-2499,2501-2524,2526-2539,2541,2543-2550,2552-2561,2564-2571,2573-2574,2576-2581,2584-2592,2594-2597
,2599-2600,2602-2607,2609-2618,2620-2628,2630-2653,2655-2662,2664-2696,2698-2703,2705-2733,2735-2738,2740-2754,2756-2769,2771-2774,2776-2800,2802-2812,2814-28
57,2859-2864,2866-2873,2875-2902,2904-2911,2914-2926,2928-2974,2977-2988,2990-2992,2994-3011,3014-3023,3025-3055,3057-3068,3070-3090,3092-3098,3100-3108,3110,
3112-3120,3122-3124,3126-3129,3131-3133,3135-3142,3144-3150,3152-3177,3179,3181-3186,3188-3202,3204-3212,3214-3220,3222-3225,3227-3241,3243-3278,3280-3281,328
3-3284,3286-3289,3291-3297,3299-3301,3303-3313,3315-3344,3346-3354,3357-3394,3396-3408,3410-3415,3417-3423,3425-3429,3431-3447,3449-3456]
boost_fua_dbg    up      10:00  3195/58/5/3258  lrdn[0001-0039,0041-0043,0045-0094,0096-0150,0152-0173,0175-0178,0180-0189,0191-0243,0245-0250,0252-0253,025
5,0258-0270,0272-0287,0289-0294,0296-0317,0319-0351,0353,0355-0360,0362,0364-0373,0375-0380,0382-0389,0391-0455,0457-0464,0466-0469,0471-0489,0491-0510,0512-0
522,0524-0538,0540-0562,0564-0621,0623-0667,0669-0670,0672-0685,0687-0709,0711-0763,0765-0829,0831-0838,0840-0884,0886-0896,0898-0936,0938-0945,0947-0969,0971
-0975,0977-0998,1000,1002-1003,1005-1012,1015,1017-1036,1039-1108,1110-1115,1117-1128,1130-1139,1141-1153,1155-1163,1165,1167-1184,1186-1187,1189-1205,1207-12
08,1210-1235,1237-1419,1421-1426,1428-1456,1458-1473,1475-1540,1542-1545,1547-1577,1579-1589,1591-1638,1640-1660,1662,1664-1668,1671-1695,1697-1709,1711-1724,
1726-1751,1753-1754,1756-1770,1772-1773,1775-1783,1785-1797,1799-1820,1822-1824,1826-1844,1846-1853,1855-1899,1901-1947,1949-1961,1963-2012,2014-2019,2021-202
7,2029-2066,2068-2070,2072-2084,2086-2102,2104-2124,2126-2138,2140-2147,2149-2170,2172-2179,2181-2217,2219-2343,2345-2348,2350-2356,2359-2362,2365-2367,2369-2
383,2385-2387,2389-2396,2398-2421,2423-2437,2439-2490,2492-2499,2501-2524,2526-2539,2541,2543-2550,2552-2561,2564-2571,2573-2574,2576-2581,2584-2592,2594-2597
,2599-2600,2602-2607,2609-2618,2620-2628,2630-2653,2655-2662,2664-2696,2698-2703,2705-2733,2735-2738,2740-2754,2756-2769,2771-2774,2776-2800,2802-2812,2814-28
```


Demo topics

Check available profiles, modules, datasets

- The software is installed by cluster admins in modules
- To list available SW **module av**
- To load a module **module load cuda**
- To load a profile **module load profile/deeplrn**
- After loading a profile you will see more SW modules available
- The deeplearning profile gives you access to common datasets (e.g. imagenet)
- You will have to load all profiles/modules required by your script each time you run a new session/job. The cluster does not know what you need beforehand.

```
[apilzer0@login07 ~]$ module av
----- /leonardo/prod/spack/03/install/0.19/linux-rhel8-icelake/gcc-8.5.0/environment-modules-5.2.0-rz47odw4phlhzhbz7b65nv5s5othgmi/modulefiles
dot module-git module-info modules null use,own

----- /leonardo/prod/opt/modulefiles/profiles -----
profile/archive profile/base profile/chem-phys profile/eng profile/lifesc profile/quantum profile/statistics
profile/astro profile/candidate profile/deeplrn profile/geo-inquire profile/meteo profile/spoke7

----- /leonardo/prod/opt/modulefiles/base/archive -----
fake/1.0

----- /leonardo/prod/opt/modulefiles/base/dependencies -----
anaconda3/2023.03

----- /leonardo/prod/opt/modulefiles/base/libraries -----
adios/1.13.1--intel-oneapi-mpi--2021.10.0--oneapi--2023.2.0 metis/5.1.0--gcc--12.2.0
adios/1.13.1--openmpi--4.1.6--gcc--12.2.0-cuda-12.1 metis/5.1.0--oneapi--2023.2.0
blitz/1.0.2--gcc--12.2.0 nccl/2.19.1--1--gcc--12.2.0-cuda-12.1
blitz/1.0.2--oneapi--2023.2.0 nccl/2.19.3--1--gcc--12.2.0-cuda-12.1
boost/1.83.0--gcc--12.2.0 netcdf-c/4.9.2--gcc--12.2.0
boost/1.83.0--intel-oneapi-mpi--2021.10.0--oneapi--2023.2.0-atomic netcdf-c/4.9.2--intel-oneapi-mpi--2021.10.0--oneapi--2023.2.0
boost/1.83.0--oneapi--2023.2.0 netcdf-c/4.9.2--oneapi--2023.2.0
boost/1.83.0--openmpi--4.1.6--gcc--12.2.0 netcdf-c/4.9.2--openmpi--4.1.6--gcc--12.2.0
boost/1.83.0--openmpi--4.1.6--nvhpc--23.11 netcdf-c/4.9.2--openmpi--4.1.6--nvhpc--23.11
cfitsio/4.3.0--gcc--12.2.0 netcdf-fortran/4.6.1--gcc--12.2.0
cgal/4.11--openmpi--4.1.6--gcc--12.2.0 netcdf-fortran/4.6.1--intel-oneapi-mpi--2021.10.0--oneapi--2023.2.0
cgal/5.5.2--gcc--12.2.0 netcdf-fortran/4.6.1--oneapi--2023.2.0
cgal/5.5.2--intel-oneapi-mpi--2021.10.0--oneapi--2023.2.0 netcdf-fortran/4.6.1--openmpi--4.1.6--gcc--12.2.0
cinca-hpyc/2023.05 netcdf-fortran/4.6.1--openmpi--4.1.6--nvhpc--23.11
cudnn/8.9.7.29-12--gcc--12.2.0-cuda-12.1 netlib-scalapack/2.2.0--openmpi--4.1.6--gcc--12.2.0
```


Demo topics

Start interactive sessions (srun, salloc)

- **srun** allows you to open a terminal or execute a script directly on the compute node
- **salloc** is allocates for you a compute node but you will still be on the login node
- To execute a code with salloc on the compute node you need to add srun before your command
- Note in the salloc example that the login node does not have gpu. It throws error when I run nvidia-smi, instead the compute node can execute it without any problem

```
[apilzer0@login07 ~]$ srun --time 00:30:00 --nodes 1 --gres=gpu:1 --ntasks=1 --cpus-per-task=8 --partition boost_usr_prod -A tra24_openhack -q boost_qos_dbg -pty /bin/bash
[apilzer0@lrdn0012 ~]$
[apilzer0@lrdn0012 ~]$ nvidia-smi
Wed May 29 17:35:02 2024

+-----+
| NVIDIA-SMI 530.30.02                | Driver Version: 530.30.02   | CUDA Version: 12.1   |
+-----+-----+
| GPU  Name                Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap|  Memory-Usage | GPU-Util  Compute M. |
|====|=====|
|  0  NVIDIA A100-SXM-64GB      On          | 00000000:08:00.0 Off |             0         |
| N/A   43C    P0              61W / 458W|  0MiB / 65536MiB |      0%    Default   |
+-----+-----+
|                                     |                               |
+-----+-----+
| Processes:                         |                               |
| GPU   GI    CI          PID    Type    Process name                  | GPU Memory |
|====|=====|
|                                     | No running processes found |
+-----+-----+
[apilzer0@lrdn0012 ~]$
```

```
[apilzer0@login07 ~]$ salloc --time 00:30:00 --nodes 1 --gres=gpu:1 --ntasks=1 --cpus-per-task=8 --partition boost_usr_prod -A tra24_openhack -q boost_qos_dbg
salloc: Granted job allocation 4786683
salloc: Waiting for resource configuration
salloc: Nodes lrdn0012 are ready for job
[apilzer0@login07 ~]$ cat nvidia-smi.sh
#!/bin/bash
nvidia-smi

[apilzer0@login07 ~]$ bash nvidia-smi.sh
nvidia-smi.sh: line 2: nvidia-smi: command not found
[apilzer0@login07 ~]$ srun bash nvidia-smi.sh
Wed May 29 17:39:28 2024

+-----+
| NVIDIA-SMI 530.30.02                | Driver Version: 530.30.02   | CUDA Version: 12.1   |
+-----+-----+
| GPU  Name                Persistence-M| Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap|  Memory-Usage | GPU-Util  Compute M. |
|====|=====|
|  0  NVIDIA A100-SXM-64GB      On          | 00000000:08:00.0 Off |             0         |
| N/A   43C    P0              61W / 458W|  0MiB / 65536MiB |      0%    Default   |
+-----+-----+
|                                     |                               |
+-----+-----+
| Processes:                         |                               |
| GPU   GI    CI          PID    Type    Process name                  | GPU Memory |
|====|=====|
|                                     | No running processes found |
+-----+-----+
[apilzer0@login07 ~]$
```


Demo topics

Submit jobs and cancel jobs (debug jobs)

- The way to run jobs on the cluster is to submit jobs to the queue with the command **sbatch your_slurm_script.sh**
- Example [here](#) for a multinode script
 - #SBATCH are instructions you give to the slurm workload manager
 - Important ones are the account to use, execution time asked, partition (CPU or GPU), number of nodes, GPUs
 - Then you should load the modules your code needs and run your scripts/executables
 - In this example you find also the definition of master address, master port, world size. These are instructions for multi-node job execution. Each job needs an orchestrator for the communications, this will be done by the master node (master address) through the master port assigned. World size means how many tasks we are running in total
- If some of these terminology is unclear please refer to the [jupyter notebook](#) where you will find more examples

Demo topics

ML experiment

- We will see a small image classification example on CIFAR-10
- A standard single GPU training defines in a nutshell a model, a dataloader, a loss (cost function) and a training loop
- A distributed training defines
 - distributed environment variables (local rank, global rank, world size)
 - Sets up distributed processes
 - wraps the model in a distributed communication class
 - distributes the dataloader
 - handles all the synchronization of the training loops running in parallel processes in the background via **nccl** protocol in the case of NVIDIA GPUs or gloo for CPUs
- Pay attention that only rank 0 prints output and saves the checkpoints to avoid concurrent prints & file writes
- Note also that we set number of workers in the dataloader to 8 because on Leonardo we have 8 CPU cores per GPU. Other clusters might be slightly different

```
if os.environ.get('OMPI_COMMAND'):  
    # from mpirun  
    rank = int(os.environ["OMPI_COMM_WORLD_RANK"])  
    local_rank = int(os.environ["OMPI_COMM_WORLD_LOCAL_RANK"])  
    world_size = int(os.environ["OMPI_COMM_WORLD_SIZE"])
```

```
# Initializes the distributed backend which will take care of synchronizing nodes/GPUs  
torch.distributed.init_process_group(backend="nccl", rank=rank, world_size=world_size)
```

```
ddp_model = torch.nn.parallel.DistributedDataParallel(model, device_ids=[local_rank], output_device=local_rank)
```

```
# Data should be prefetched  
# Download should be set to be False, because it is not multiprocessing safe  
train_set = torchvision.datasets.CIFAR10(root="data/cifar-10", train=True, download=True, transform=transform)  
test_set = torchvision.datasets.CIFAR10(root="data/cifar-10", train=False, download=True, transform=transform)  
  
# Restricts data loading to a subset of the dataset exclusive to the current process  
train_sampler = DistributedSampler(dataset=train_set)
```

```
train_loader = DataLoader(dataset=train_set, batch_size=batch_size, sampler=train_sampler, num_workers=num_workers)  
# Test loader does not have to follow distributed sampling strategy  
test_loader = DataLoader(dataset=test_set, batch_size=128, shuffle=False, num_workers=num_workers)
```

```
Epoch: 1, Accuracy: 0.1546  
-----  
Rank: 0, Epoch: 1, Training ...  
Time 11.23 seconds  
-----  
Epoch: 2, Accuracy: 0.3986  
-----  
Rank: 0, Epoch: 2, Training ...  
Time 10.85 seconds  
-----  
Epoch: 3, Accuracy: 0.4261  
-----  
Rank: 0, Epoch: 3, Training ...  
Time 10.69 seconds  
-----  
Epoch: 4, Accuracy: 0.4787  
-----  
Rank: 0, Epoch: 4, Training ...  
Time 10.98 seconds  
-----
```